SIM Phase-Control Dataset — Report

Classical-optimizer-generated training data for the Transformer controller

1. Purpose

Before training a Transformer to control the SIM phase shifts, we need a dataset that pairs wireless channels with the best known SIM configuration for each one. This dataset is generated using a classical (non-AI) optimization method, which acts as the "teacher" the Transformer will later learn to imitate, and eventually improve on.

2. How the Channels Are Generated

Each channel is a Rician fading channel: a combination of a strong, deterministic line-of-sight component and a random scattered (Rayleigh) component, scaled by the physical path loss between the SIM and the user, and spatially correlated across neighboring metasurface elements. The user position is fixed; only the random scattering component changes from one sample to the next, so the dataset captures variability in instantaneous fading, at a single fixed distance/geometry.

3. How the Optimal Phases Are Found

For each channel, the SIM phase shifts are optimized with gradient ascent: starting from a random configuration, the phases are repeatedly adjusted in the direction that increases the user's achievable rate, with a backtracking line search so that every accepted step actually improves the rate. The optimizer stops automatically once no further improvement is found. The final phases are then quantized to match the real hardware constraint (2-bit phase shifters, 4 discrete values).

4. Configuration Used

Parameter	Value	Meaning
K (users)	1	Single-user validation stage, as recommended
L (SIM layers)	5	Number of metasurface layers
N (elements/layer)	49	7x7 meta-atom grid per layer
NbitQ	2 bits	4 possible phase values per element
Rician K-factor	10	Strong line-of-sight component
Samples (M)	2000	Number of (channel, phase) pairs generated

5. What the Dataset File Contains

- **H** (2000 x 49, complex) — the channel samples.
- **theta** (2000 x 245, real) — the matching optimal, quantized phase shifts (the Transformer's training target).
- **rate** (2000,) — the achievable rate obtained with each theta.
- **convergence_curves** — a few saved optimization trajectories, kept only to visually verify the optimizer converges correctly.
- **metadata** — every parameter used to generate the dataset, so it can be reproduced exactly.

6. Validation — Why We Trust This Dataset

- **Convergence check**: for several sample channels, the rate increases smoothly with each optimization step and then flattens out, confirming the optimizer reaches a real maximum rather than stopping early or diverging.
- **Rate distribution**: the 2000 achieved rates form a clean, realistic bell-shaped distribution (mean about 8.2 bit/s/Hz), with no broken values, zeros, or extreme outliers.
- **Channel magnitude check**: the raw channel values are small (around 0.0001 in magnitude), exactly matching the expected physical path loss at the simulated distance and frequency - confirming the channel model itself is physically correct, not just numerically stable.

7. Known Limitations (For Next Steps)

- Only one user distance/geometry is used; the random part is only the fading, not the user's position. Later datasets should also vary distance and/or the Rician factor for broader generalization.
- Single-user only (K=1); multi-user (K>1, with interference) is the next stage, as planned.

8. Bugs Fixed Along the Way

Three pre-existing bugs in the shared physics code were found and corrected while building this dataset: a missing path-loss scaling and a shape error in the Rician channel generator, and a crash in the classical optimizer that only appeared in the single-user (K=1) case. All three are now fixed for the whole codebase.